

Laboratory Activity No. 1

Real-Time Multisensor Room Monitoring System — Technical Laboratory Report

PlatformIO | STM32Cube | STM32F103C8 Blue Pill | FreeRTOS | Wokwi

1. Problem and Requirements

For this laboratory, I built a simulated room monitor that runs on an STM32 Blue Pill. The device samples temperature, humidity, and relative light; detects motion; shows one selected measurement on an OLED; sounds an alarm whenever the temperature leaves its normal range; and drops into an inactive state after a stretch with no motion.

The required temperature limits are 18 °C and 30 °C, and both endpoints count as normal — the alarm only fires low below 18 °C and high above 30 °C. The implementation is built on the STM32Cube framework with native FreeRTOS APIs, not the Arduino framework.

Wokwi translates the physical system into a virtual Blue Pill circuit with a DHT22, a photoresistor module, a PIR sensor, a KY-040 encoder, an SSD1306 OLED, and a buzzer. The LDR output is expressed as a relative percentage rather than calibrated lux.

One boundary worth stating up front: the DHT22 is wired up in the Wokwi diagram and the firmware now communicates with it through a bounded single-wire driver on PA1. The driver rejects invalid frames, and the report distinguishes implemented acquisition from runtime Wokwi verification.

2. System Architecture and Design

Hardware architecture

Device	Pin / interface	Role
STM32F103C8 Blue Pill	MCU; internal HSI clock configured at 8 MHz	Runs application tasks and FreeRTOS

Device	Pin / interface	Role
Photoresistor module	PA0, ADC1 channel 0	Relative ambient light input
DHT22	PA1 in the circuit	Temperature and humidity input through the single-wire driver with checksum/range validation
PIR sensor	PA2	Motion input
KY-040 encoder	PA3 CLK, PA4 DT, PA5 switch	Rotary page selection; firmware samples CLK and DT, but does not read the switch
SSD1306 OLED	PB6 SCL, PB7 SDA, I2C address 0x3C	Displays the selected page
Buzzer	PA8, TIM1 channel 1 PWM	Temperature alarm output
Serial monitor	PA9 TX, PA10 RX, USART1 at 115200 baud	Startup and task diagnostics

The MCU clock setup uses the internal 8 MHz HSI without a PLL. The HAL millisecond tick runs on TIM4 because the FreeRTOS port owns SysTick. The OLED communicates over a project-local bit-banged I2C implementation on PB6/PB7, and the buzzer is set up for a nominal 1 kHz PWM signal.

Software architecture

The application is split across sensor, display, input, alarm, motion, system-state, and RTOS-object modules. The pure decision functions for temperature alarms, encoder navigation, and system-state transitions are kept apart from the peripheral access code so they can be compiled and tested natively.

```

SensorTask --+--> xQueueSensorToDisplay --> DisplayTask --> SSD1306
              +--> xQueueSensorToAlarm   --> AlarmTask   --> buzzer
InputTask  -----> xQueueInputToDisplay  --> DisplayTask
MotionTask -----> EVENT_MOTION          -----> StateTask
StateTask  -----> EVENT_ACTIVE          -----> other tasks
AlarmTask  -----> EVENT_ALARM            -----> DisplayTask
All task diagnostics -----> USART1, guarded by serialMutex

```

DisplayTask is the only task that touches the OLED at runtime. SensorTask publishes each sensor record to separate display and alarm queues because a queue receive consumes the item. The sensor queues each hold five SensorData records; the mode queue holds four DisplayMode values. The event group bits are EVENT_ACTIVE (StateTask owner), EVENT_MOTION (MotionTask owner), and EVENT_ALARM (AlarmTask owner). The USART1 mutex is named serialMutex in code; it protects each transmit call, not necessarily a whole printf record.

State machine

```
Boot -> ACTIVE
ACTIVE -- 15 seconds without motion --> INACTIVE
INACTIVE -- motion detected -----> ACTIVE
```

The system boots into ACTIVE. StateTask owns EVENT_ACTIVE and applies the 15-second inactivity rule. While the system is inactive, MotionTask keeps sampling the PIR so any activity can wake it up; SensorTask skips acquisition; InputTask keeps polling but throws away mode changes; DisplayTask blanks the screen the next time it checks the state; and AlarmTask silences the buzzer.

3. FreeRTOS Architecture

All six application tasks are created in main.cpp with explicit priorities. The periodic input, motion, and sensor work uses vTaskDelayUntil() so each run stays aligned to a fixed period instead of letting the delay accumulate after every task body. Tasks waiting on queues or event groups sit in the Blocked state until their data, event, or timeout arrives.

Task	Responsibility	Trigger / period	Priority	IPC	Typical blocked condition
SensorTask	Read LDR and current temperature/humidity values; publish SensorData to two consumer queues	2 s, vTaskDelayUntil()	2	Two sensor queues; reads EVENT_ACTIVE and EVENT_MOTION	Periodic delay; queue send can wait up to 50 ms

Task	Responsibility	Trigger / period	Priority	IPC	Typical blocked condition
DisplayTask	Render the selected measurement; sole OLED owner	Sensor update or queue timeout	1	Sensor queue and mode queue	Waits up to 2.5 s for a sensor queue item
InputTask	Decode encoder CLK/DT and select a display page while ACTIVE	5 ms, vTaskDelayUntil()	3	Display-mode queue; reads EVENT_ACTIVE	5 ms periodic delay
MotionTask	Sample the PIR and mirror its level to EVENT_MOTION	100 ms, vTaskDelayUntil()	3	System event group	100 ms periodic delay
AlarmTask	Evaluate temperature, drive buzzer PWM, publish EVENT_ALARM	Sensor data; 500 ms queue timeout	2	Alarm sensor queue and system event group	Waits up to 500 ms for sensor data
StateTask	Apply ACTIVE/INACTIVE transitions and inactivity timeout	Motion event or 15 s timeout	2	Waits on EVENT_MOTION; owns EVENT_ACTIVE	Event wait; 500 ms delay after detecting asserted motion

The priorities follow the latency requirements. Encoder and PIR input run at priority 3 because those signals can change on a short timescale. Sensor, alarm, and state processing sit at priority 2. OLED updates run at priority 1, since a small delay on the display is acceptable and a slow display operation shouldn't outrank input or sampling work.

In FreeRTOS terms, a task that is eligible to run but waiting for the CPU is Ready; a task waiting on a delay, queue item, or event is Blocked. The periodic tasks finish their finite work and return to Blocked. All tasks and RTOS objects are created dynamically from the configured FreeRTOS heap.

4. Implementation

SensorData holds temperature, humidity, lightLevel, and motionDetected. DHT22_Read() captures the PA1 single-wire frame with bounded DWT timing; dht22Decode() checks the checksum and value range before SensorTask publishes a sample. A failed DHT22 read is logged and skipped. The LDR is sampled through ADC1 channel 0 on PA0 and lightPercentFromAdc() scales the result to 0-100% relative brightness, not calibrated lux; an ADC failure retains the last value.

DisplayMode is an enumeration with TEMPERATURE, HUMIDITY, LIGHT, and MOTION values. InputTask watches for a falling edge on encoder CLK and compares DT to decide between forward and reverse navigation, then posts the new mode to a queue; DisplayTask performs the actual OLED update. The encoder switch input is configured on PA5 but isn't processed yet.

Temperature evaluation is deliberately hardware-independent: values below 18 °C map to LOW_TEMPERATURE, values above 30 °C map to HIGH_TEMPERATURE, and everything from 18 °C through 30 °C maps to NORMAL. AlarmTask drives TIM1 channel 1 PWM on PA8 whenever the system is ACTIVE and the temperature is out of range.

The serialMutex is taken inside the _write() retarget around each HAL_UART_Transmit() call once the scheduler has started; startup output is emitted before the scheduler runs. It protects the USART1 peripheral transmit call, not necessarily an entire printf call as one indivisible record. The symbol in this project is serialMutex, not xSerialMutex.

5. Verification and Testing

Build and unit tests

Verification	Recorded result	Scope / caveat
pio run	PASS; 50.2% RAM and 51.9% flash	Default bluepill_f103c8 environment builds successfully
pio test	ERROR; zero tests executed	Default STM32Cube environment has no Unity configuration; use the native environment
pio test -e native	PASS; 26/26 tests, 0 failed	5 alarm, 4 navigation, 3 encoder, 5 DHT22 decode, 4 light-scaling, and 5 state cases; no hardware integration tests

Verification	Recorded result	Scope / caveat
pio check	PASS; 0 high, 0 medium, 20 low findings	Runs cppcheck on default bluepill_f103c8; details and dispositions below
pio check -e native	PASS; no defects found	Toolchain-define extraction warnings were nonfatal

Wokwi functional verification

The current-firmware terminal output showed samples every 2 seconds with 24.00 C, 40.00% humidity, 76% light, and no motion. After approximately 15 seconds without motion it printed `STATE: INACTIVE` (no motion for 15 s). This is evidence of live sampling and the inactivity transition. It does not show that the sensor controls were changed or that encoder, alarm, PIR activation, OLED blanking, and wake-up behavior were exercised. The supplied older screenshot has different values and is not evidence for the current firmware revision.

Test	Laboratory stimulus	Observed behavior	Result
FT-01	Change temperature	Sample reported 24.00 C; a controlled temperature change was not recorded	Partial
FT-02	Change humidity	Sample reported 40.00% RH; a controlled humidity change was not recorded	Partial
FT-03	Change light input	Sample reported 76%; a dark/bright comparison was not recorded	Partial
FT-04	Rotate encoder clockwise	Not observed in the supplied current-firmware log	Pending
FT-05	Rotate encoder counterclockwise	Not observed in the supplied current-firmware log	Pending
FT-06	Set temperature above 30 C	Alarm/buzzer response not observed	Pending

Test	Laboratory stimulus	Observed behavior	Result
FT-07	Return temperature to normal	Alarm/buzzer stop behavior not observed	Pending
FT-08	Trigger PIR while ACTIVE	Sample showed no motion; a PIR trigger was not recorded	Pending
FT-09	Allow 15 seconds without motion	Terminal printed STATE: INACTIVE (no motion for 15 s) after the no-motion interval	PASS (reported observation; no stopwatch value recorded)
FT-10	Trigger PIR while INACTIVE	Wake-up, OLED resume, and sampling resume were not observed	Pending

Requirements traceability

This matrix follows the laboratory PDF's FT identifiers. Code implementation and automated tests are distinguished from runtime verification; only the observed FT-09 state transition is marked PASS.

Requirement	Implementation / task	Unit verification	Wokwi verification / status
FR-01 Temperature	SensorTask; DHT22_Read() / dht22Decode()	test_dht22_decodes_valid_frame; checksum/range tests	FT-01 partial: 24.00 C sample; controlled change pending
FR-02 Humidity	SensorTask; DHT22_Read() / dht22Decode()	test_dht22_decodes_valid_frame; checksum/range tests	FT-02 partial: 40.00% sample; controlled change pending

Requirement	Implementation / task	Unit verification	Wokwi verification / status
FR-03 Light	SensorTask; lightPercentFromAdc() on ADC1 PA0	Four test_light_adc_* cases	FT-03 partial: 76% sample; dark/bright comparison pending
FR-04 Motion	MotionTask samples PA2 and sets/clears EVENT_MOTION	State logic unit tests exercise motion inputs	FT-08 pending: PIR trigger not recorded
FR-05 OLED	DisplayTask exclusively owns SSD1306 and renders one DisplayMode	Display logic exercised by navigation decisions; no OLED integration test	FT-01 to FT- 03 display updates not verified on current run
FR-06 Encoder navigation	InputTask; encoderStep(), nextDisplayMode(), previousDisplayMode()	Four test_navigation_* and three test_encoder_* cases	FT-04/FT-05 pending: turns not recorded
FR-07 Temperature alarm	AlarmTask; evaluateTemperature(); TIM1 PWM buzzer	Five test_alarm_* boundary cases	FT-06/FT-07 pending: buzzer response not recorded
FR-08 Activity state	StateTask/MotionTask; evaluateSystemState()	Five test_state_* cases	FT-08 pending: ACTIVE motion behavior not recorded

Requirement	Implementation / task	Unit verification	Wokwi verification / status
FR-09 Auto inactivity	StateTask; kInactivityTimeoutMs = 15000	test_state_active_timeout_becomes_inactive	FT-09 PASS: INACTIVE message observed after no-motion interval
FR-10 Auto reactivation	MotionTask EVENT_MOTION plus StateTask ACTIVE restore	test_state_inactive_motion_reactivates; test_state_motion_wins_over_timeout	FT-10 pending: PIR wake-up not recorded

6. Static Code Analysis

I ran PlatformIO's cppcheck for both the firmware and native environments. The firmware environment reported no high- or medium-severity findings and 20 low-severity findings, while the native logic-only check reported no defects. The low findings are documented in docs/static-analysis.md and consist primarily of STM32 register-macro casts plus a bounded-timeout diagnostic.

The native check intentionally skips STM32-only sources such as the custom port, so its clean result applies to the hardware-independent decision logic rather than the complete firmware source set. PlatformIO also emitted some nonfatal toolchain-define extraction warnings during the native check.

File / line	Finding / cause	Resolution
src/dht22.cpp: 33, 39, 63, 69, 80, 81, 82, 84, 87	Low style C-style pointer casts reported through CMSIS GPIO/DWT register macros	Accepted vendor/register diagnostics; changing CMSIS register access would require altering framework headers
src/dht22.cpp: 39	Low unsignedLessThanZero diagnostic on the timeout bound	Timeout remains bounded by an iteration guard and elapsed DWT cycles; diagnostic

File / line	Finding / cause	Resolution
		is a false positive for the unsigned counter
src/main.cpp: 41, 53, 54, 55, 56, 57, 66, 67, 198	Low style C-style pointer casts in FreeRTOS/CMSIS register macros and vector-table setup	Accepted framework/register access patterns
src/main.cpp: 150	Low constParameterPointer suggestion for HAL_UART_Mspltinit callback parameter	Retain the HAL-required callback signature

7. Engineering Discussion

The main architectural trade-off here is splitting sensor fan-out across two queues. It avoids a race in which DisplayTask and AlarmTask would otherwise compete to consume the same queue item, at the cost of publishing the same record twice. I used the event group for shared state flags because several tasks need to read the current ACTIVE and MOTION levels, and a mutex on the UART because several task contexts emit diagnostics.

One consequence of the design: the display task waits up to 2.5 seconds on its sensor queue before checking system activity and redrawing, so a mode change or an inactive transition may not show up immediately. InputTask keeps polling while inactive but discards mode changes, and the PA5 encoder switch isn't consumed. These are known limitations, not verified feature claims.

The project uses a custom FreeRTOS Cortex-M3 port, chosen for how it behaves in the Wokwi simulator — a simulator-specific engineering decision, since physical hardware testing hasn't been documented. The DHT22 protocol is now implemented with bounded timing and checksum validation; its temperature, humidity, and alarm behavior still require a recorded Wokwi run.

8. Final Submission Checklist

This checklist applies Part XXV of the laboratory guide to the current repository. Complete means that the implementation or documentation is present and supported by recorded evidence; pending means that the item still requires an observed result or publication step.

Category	Requirement	Evidence in this project	Status
Build	pio run succeeds	Recorded firmware build passes for bluepill_f103c8	Complete
FreeRTOS	At least five meaningful tasks	Six application tasks are created and perform separate responsibilities	Complete
FreeRTOS	Explicit priorities	Priorities are assigned in main.cpp and justified in the report	Complete
FreeRTOS	Queue	Sensor, alarm, and display-mode queues carry application data	Complete
FreeRTOS	Mutex	serialMutex protects shared USART1 transmission	Complete
FreeRTOS	Event group or task notification	xSystemEvents carries ACTIVE, MOTION, and ALARM bits	Complete
FreeRTOS	vTaskDelayUntil() used appropriately	SensorTask, InputTask, and MotionTask use periodic blocking delays	Complete
FreeRTOS	ACTIVE/INACTIVE state machine	StateTask applies the 15-second timeout and motion reactivation	Implemented; runtime pending
Simulation	DHT22, LDR, PIR, rotary encoder, OLED, and buzzer	All required parts are present in diagram.json; DHT22 acquisition is implemented on PA1	Implemented; runtime pending
Simulation	Wokwi simulation operational	Build and recorded OLED/serial evidence exist; complete interaction record is pending	Partial
Testing	At least 13 meaningful unit tests	26 native tests cover alarm, navigation/encoder, DHT22 decoding, light scaling, and state decisions	Complete
Testing	Unit tests pass	pio test -e native: 26/26 pass; plain pio test selects STM32Cube and fails to find Unity configuration	Partial: native suite passes; default command fails

Category	Requirement	Evidence in this project	Status
Testing	Functional verification table completed	FT-01 through FT-10 are documented, but observations remain pending	Pending
Testing	Fault experiments completed	Procedures and expected observations are documented in fault-experiments.md; results are not recorded	Pending
Quality	pio check completed	Firmware and native cppcheck results are recorded	Complete
Quality	Findings analyzed and significant warnings addressed	High/medium findings are zero; 20 low findings are interpreted in docs/static-analysis.md	Complete
GitHub	Public repository	Project is published at github.com/Mahather11/bca182-freertos-multisensor	Complete
GitHub	Meaningful commit history	Development milestones are recorded in Git history and development-log.md	Complete
GitHub	Professional README	README documents architecture, build, testing, limitations, and verification scope	Complete
GitHub	Architecture diagrams	README includes system and state-machine diagrams	Complete
GitHub	No unnecessary binaries/build artifacts	Required laboratory report PDF is committed; repository artifact review remains advisable	Review
Report	Laboratory report included	docs/laboratory-report.pdf and its HTML source are present	Complete
Report	FreeRTOS task table	Section 3 documents tasks, priorities, periods, IPC, and blocked conditions	Complete
Report	Requirements traceability matrix	FR-01 through FR-10 are mapped to implementation and evidence	Complete

Category	Requirement	Evidence in this project	Status
Report	Test results and static-analysis discussion	Sections 5 and 6 record native tests, Wokwi limits, and cppcheck findings	Complete
Report	Limitations documented	Pending Wokwi verification and unrecorded fault experiments are stated	Complete
Portfolio	Hackster.io article	No Hackster.io publication is recorded in this repository	Pending
Portfolio	GitHub link and attribution included	Repository is public and the project identity is documented	Partial

9. Conclusion

All told, the project delivers a modular, concurrent STM32 FreeRTOS application: six distinct tasks, queues, an event group, a serial mutex, an OLED display path, PIR-based state logic, and independently testable alarm, navigation, sensor-decoding, light-scaling, and state decisions. The recorded firmware build and 26 native logic tests pass.

That said, it isn't yet a complete functional demonstration of every lab requirement. The DHT22 driver is implemented, but controlled sensor changes, the remaining FT-01 through FT-10 interactions, and the fault-experiment observations are not fully recorded. The next engineering steps are to run those simulator tests against a recorded firmware revision and capture current-revision screenshots.